Towards AI Swarms Detection: A Conceptual Paper on Emergent, Early Warning for P2P AI Agents

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Abstract

We propose a conceptual, tiered framework for detecting *AI swarms*: collections of distributed, interacting AI agents (for example, LLM-powered assistants running on personal devices) that manifest emergent, coordinated behavior outside controlled data centers. Our primary observable is network communication patterns, specifically inter-message intervals and message characteristics from P2P traffic, which autonomous agents generate at sufficient frequency to enable statistical characterization within minutes. This manuscript is a position and technical-foundations paper: we do not claim evaluation on a real-world dataset (none public is available for the specific use-case studied), but instead introduce the definitions, physical and statistical analogies, mathematical primitives, and a multi-tiered detection pipeline inspired by seismographic early-warning systems. We argue that early detection requires continuous, multi-scale monitoring of behavioral anomaly signals and their spatial/temporal coherence; present a mathematically explicit detector that measures anomaly magnitude and multi-scale coherence; discuss risks, limitations, and tuning; and include an appendix with synthetic scenario results and parameters. The goal is to provide a rigorous blueprint for future empirical work and responsible deployment.

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1 Introduction

Large language models (LLMs) and their derivatives are increasingly embedded in personal devices and edge applications. As LLM-powered agents proliferate, a new class of threats and unanticipated behaviors can arise: ensembles of independently-operated assistants interacting in an environment (directly, or indirectly via shared services) that produce *collective* or *emergent* behaviors. Such phenomena is an almost direct consequence of the design of so-called *AI swarms* which gained rapid popularity in 2025[7].

This paper is a position piece that (1) articulates threat scenarios related to AI swarms, (2) describes the conceptual analogy to seismographs and early-warning detection, (3) develops mathematical primitives to quantify anomaly magnitude and spatial coherence, and (4) proposes a tiered detection architecture that balances sensitivity, interpretability, and deployability. We do not present an evaluation on production P2P traffic; instead we provide synthetic experiments and detailed parameter justifications in the appendix to illustrate behavior and to support empirical future work.

The experiments were run on a simulated pipeline used to collect the results presented in appendix. The source code of our pipeline is shared on GitHub and Kaggle[6, 5].

1.1 Scope and disclaimers

This manuscript intentionally focuses on detection and not on offensive capabilities. It is a conceptual and mathematical framework intended to guide future empirical collection, evaluation, and responsible deployment. We do not rely on any privileged dataset and we highlight limitations and ethical considerations throughout.

2 What is an AI swarm?

We define an **AI swarm** as a collection of autonomous or semi-autonomous AI agents operating across multiple devices and/or network nodes whose interactions (direct or mediated) produce *coordinated, emergent behavior* that is *distinct* from the behavior of the same agents operating independently. Key properties:

- **Distribution**: agents are geographically and topologically distributed (mobile devices, P2P networks, IoT).
- **Local autonomy**: each agent acts on local observations and policies (possibly influenced by shared models, prompts, or incentives).
- **Interaction**: agents exchange information (direct P2P, shared server, social signals), intentionally or as a byproduct of environment interaction.
- **Emergence**: coordination arises via dynamics and interactions rather than via a single centralized controller.
- **Adaptive / temporal dynamics**: patterns evolve over time, potentially rapidly.

2.1 Risks and safety concerns

AI swarms pose a range of potential security and safety risks beyond single-device[8, 9] AI threats. Examples include:

- **Coordinated misinformation or influence campaigns** that propagate across social meshes with LLM-crafted content optimized to evade filters.
- **Resource harassment** (e.g. coordinated deluge of queries or messages causing local service degradation).
- **Physical safety risks** when swarms coordinate actions in the physical world (e.g., many devices controlling actuators or interacting with traffic systems).
- **Privacy erosion** via correlated probing of sensors across devices to infer private states.
- **Evasion of centralized monitoring** by leveraging P2P channels and local agent autonomy.

These risks are amplified compared to traditional botnets because modern agents have richer semantic capabilities and can adapt language and behaviors to avoid simple heuristics.

3 Motivation for multi-scale monitoring

Seismographs and seismic early-warning systems detect small local perturbations and use spatial coherence and timing to warn of larger events. The key insight we borrow is that emergent coordination typically leaves *multi-scale* signatures: local anomalies that strengthen when aggregated across space and/or time. Detection therefore benefits from:

1. a sensitive local anomaly detector (analogous to a seismometer),
2. spatial weighting to build localized fields (analogous to arrays of sensors),
3. hierarchical aggregation (analogous to multi-scale waveform analysis) to detect persistence and scaling behavior.

From this analogy we motivate a tiered detection pipeline: early per-agent anomaly scoring (Tier 1), spatial and short-time correlation checks (Tier 2), a localized field representation (Tier 2.5), and multi-scale coarse-graining to measure magnetization-like coherence (Tier 3).

4 Mathematical primitives

We rely on a set of classical mathematical concepts from statistics, spatial analysis, and statistical physics. We briefly recap the relevant background.

4.1 Heavy-tailedness and tail exponent α

The tail index α of a distribution (e.g., an α-stable model) measures heaviness of tails. When $0 < \alpha \leq 2$, heavier tails (small α) indicate more frequent large deviations. Estimation methods include Hill estimators and other robust procedures. Heavy tails have been used historically to detect extreme events in finance and network traffic.

4.2 Kurtosis κ

Kurtosis (fourth standardized moment) measures tail-weight and propensity for outliers. While sample kurtosis has known sensitivities, it is a classical and fast-to-compute indicator of anomalous burstiness.

4.3 Hurst exponent H

The Hurst exponent quantifies long-range dependence and persistence in time series (originally studied by H. E. Hurst in hydrology). Values $H > 0.5$ indicate persistence; $H \approx 0.5$ is memoryless; $H < 0.5$ indicates anti-persistence.

4.4 Spatial autocorrelation and Moran's I

Moran's I is a classical statistic for spatial autocorrelation (Moran, 1950): given observations x_i at locations, Moran's I measures similarity among neighboring nodes. It is related to the idea of a field of spins in statistical physics.

While similarity could be measured with other techniques (like Getis-Ord statistics), the Moran-based field creates a continuous, scale-ready representation: this field is a suitable input for a hierarchy of block sizes.

4.5 Scaling and block-spin

Kadanoff's block-spin conceptualization and later renormalization group (RG) methods formalize how local interactions aggregate into macroscopic behavior. Aggregating local variables into blocks and measuring how order parameters change with scale is a central idea in physics that we adapt for swarm detection.

See Section 9 for a detailed discussion of the historical roots and synthesis of these ideas

5 Tiered detection architecture: overview

We present four conceptual tiers. Each tier has increasing computational complexity and decreasing immediacy, designed to allow staged alarms and human-in-the-loop verification.

5.1 Tier 1 — Per-agent anomaly scoring (local)

Goal: compute a compact, normalized anomaly score $A \in [0, 1]$ per agent from short-term behavioral metrics.

Inputs (network observables): Computed from sliding windows of recent message traffic:

- Empirical tail index $\hat{\alpha}$ (estimated via Hill estimator on IMI or message size distributions),
- Sample kurtosis $\hat{\kappa}$ of IMI sequence,
- Estimated Hurst exponent $\hat{H}$ on IMI time series (via DFA or R/S analysis).

Canonical formula: Let $\hat{\alpha}, \hat{\kappa}, \hat{H}$ be the normalized metrics. A simple composite anomaly score is:

$$A \;=\; w_\alpha \cdot f_\alpha(\hat{\alpha}) \;+\; w_\kappa \cdot f_\kappa(\hat{\kappa}) \;+\; w_H \cdot f_H(\hat{H}), \tag{1}$$

where $w_\cdot \geq 0$ and $w_\alpha + w_\kappa + w_H = 1$ are weights, and each f maps the raw metric to $[0, 1]$ with higher values indicating greater anomaly. For instance:

$$f_\alpha(\hat{\alpha}) = \mathrm{clip}\left(\frac{\alpha_0 - \hat{\alpha}}{\alpha_0 - \alpha_{\min}}, 0, 1\right), \tag{2}$$

$$f_\kappa(\hat{\kappa}) = \mathrm{clip}\left(\frac{\hat{\kappa} - \kappa_0}{\kappa_{\max} - \kappa_0}, 0, 1\right), \tag{3}$$

$$f_H(\hat{H}) = \mathrm{clip}\left(\frac{\hat{H} - H_0}{H_{\max} - H_0}, 0, 1\right). \tag{4}$$

Interpretation: lower-than-normal α, higher kurtosis, or higher Hurst increase anomaly.

Recommended weights. Based on synthetic experiments (Appendix):

$$w_\alpha = 0.3, \quad w_\kappa = 0.5, \quad w_H = 0.2,$$

prioritizing kurtosis while balancing complementary signals. These should be tuned on operational data.

5.2 Tier 2 — Spatial correlation

Goal: check whether high anomaly scores cluster spatially (or in a network topology) more than expected by chance.

Approach: compute local weighted averages and Moran-style statistics. For agents indexed by i at positions $\mathbf{r}_i$, define a spatial weight $w_{ij} = W(\mathbf{r}_i, \mathbf{r}_j)$ (e.g., Gaussian kernel):

$$W(\mathbf{r}_i, \mathbf{r}_j) = \exp\left(-\frac{\|\mathbf{r}_i - \mathbf{r}_j\|^2}{2\sigma^2}\right) \cdot \mathbf{1}_{\{\|\mathbf{r}_i - \mathbf{r}_j\| \le d_{\max}\}}. \tag{5}$$

Define the local field

$$L_i = \frac{\sum_j w_{ij}\, s_j}{\sum_j w_{ij}} \quad \text{where } s_j \equiv \phi(A_j) \tag{6}$$

and ϕ maps anomaly score to a local "spin" or tendency variable (discussed below). Moran's I and local variants can be used:

$$I = \frac{N}{S_0} \frac{\sum_i \sum_j w_{ij}(s_i - \bar{s})(s_j - \bar{s})}{\sum_i (s_i - \bar{s})^2}, \quad S_0 = \sum_i \sum_j w_{ij}. \tag{7}$$

Interpretation: positive local Moran means neighboring agents have similar anomaly tendencies; negative means dispersion.

5.3 Tier 2.5 — Field representation ("Moran balls")

Goal: produce a spatially-smooth field of *local spin* values that preserve neighborhood structure for subsequent coarse-graining.

Mapping anomaly to spin. We recommend a continuous mapping $\phi : [0, 1] \to [-1, 1]$ so that small differences matter and hard thresholds are avoided. A convenient choice:

$$\phi(A) \;=\; \tanh\big(\gamma\,(A_0 - A)\big), \tag{8}$$

where parameters $\gamma > 0$ (gain) and A_0 (midpoint) set separation. This maps small anomalies to positive spins ("normal") and large anomalies to negative spins ("anomalous") continuously.

Computation. Using weights w_{ij},

$$\ell_i \;=\; \frac{\sum_j w_{ij}\, \phi(A_j)}{\sum_j w_{ij}}. \tag{9}$$

We call $\{\ell_i\}$ the *Moran ball field.* Each entry is a local, smoothed spin.

5.4 Tier 3 — Multi-scale coarse-graining and magnetization

Goal: detect multi-scale coherence (a macroscopic order parameter), analogous to magnetization in spin systems, and use scale dependence to classify emerging swarms.

Block aggregation (Kadanoff-style). Partition the domain into spatial blocks (grid cells) at scale s (block size). For a block B, define block magnetization

$$m_B(s) = \frac{1}{|B|} \sum_{i \in B} \ell_i \in [-1, 1]. \tag{10}$$

Collect $\{m_B(s)\}$ over all blocks at that scale.

Deviation and coherence. To robustly measure a macroscopic anomaly we combine two ingredients:

- **Deviation from ambient** $d(s) = \mathbb{E}_B\big[|m_B(s) - m_{\text{ambient}}|\big]/2 \in [0, 1]$; here m_{ambient} is the expected normal spin (usually close to $+1$).
- **Inter-block coherence** $c(s) = \frac{1}{1+\mathrm{CV}(\{m_B(s)\})} \in (0, 1]$, where CV is the (weighted) coefficient of variation.

Combine multiplicatively:

$$M(s) = d(s) \cdot c(s) \in [0, 1]. \tag{11}$$

$M(s)$ is the magnetization at scale s: large values mean blocks are both *far from normal* and *mutually coherent*.

Scale-profile and slope. Compute $M(s)$ for a set of scales $s_0 < s_1 < \cdots < s_L$ and analyze its log-profile. Define

$$\tilde{M}(k) = \log\big(M(s_k) + \epsilon\big), \qquad k = 0, \ldots, L, \tag{12}$$

and fit a slope β to $\tilde{M}(k)$ vs. k. Interpretation:

- $\beta > 0$: coherence *strengthens* with scale (hierarchical or directional swarm).
- $\beta \approx 0$: scale-persistent coherence (global coordination).
- $\beta < 0$: coherence decays with scale (local, transient clusters).

5.5 Classification rule (conceptual)

We use three descriptors: $M_{\text{fine}} \equiv M(s_0)$ (fine scale), $M_{\text{primary}} \equiv \max_k M(s_k)$, and slope β. A simple conceptual classifier:

- **LOW risk**: M_{primary} small (e.g. < 0.25) and M_{fine} small.
- **MEDIUM risk**: moderate M_{primary} (e.g. 0.25–0.45) or conflicting slope signals.
- **HIGH risk**: $M_{\text{primary}} > 0.45$ and β non-negative (persistent or strengthening coherence).

These thresholds are conceptual and must be tuned on realistic datasets.

6 Observable specification and measurement protocol

A detection framework is only as practical as its observables are measurable. We now explicitly specify what behavioral data our system monitors and justify its feasibility for autonomous AI agents.

6.1 Primary observable: network communication patterns

Our framework monitors **peer-to-peer network traffic patterns** generated by autonomous AI agents. Specifically, we extract statistical features from:

- **Inter-message intervals (IMI)**: Time gaps $\{\Delta t_k\}$ between consecutive messages sent by an agent.
- **Message sizes**: Byte counts $\{b_k\}$ of transmitted packets.
- **Communication graph features**: Frequency and topology of peer connections (optional enrichment).

Why network patterns? Network communication is the *observable consequence* of coordination. Unlike internal decision processes (which require invasive instrumentation), message timing and volume are:

- **Externally observable**: Can be monitored at network layer without accessing agent internals.
- **Privacy-preserving**: Statistical features (tail index, kurtosis) aggregate over content, avoiding deep packet inspection.
- **High-frequency**: Autonomous agents generate sufficient messages for statistical characterization.

6.2 Sampling frequency and time-to-detection: our assumptions

Autonomous agent communication rates. Modern autonomous AI agents (e.g., P2P coordinators, edge assistants with agentic capabilities could generate:

- **Low activity**: 10 messages/minute (polling, keep-alive, occasional coordination)
- **Moderate activity**: 30–50 messages/minute (active peer discovery, data exchange)
- **High activity**: 100+ messages/minute (coordinated actions, swarm behavior)

Sample accumulation timeline. Statistical estimators require varying sample sizes for reliability:

- **Kurtosis** κ: 30–50 samples $\rightarrow$ 30 sec to 5 min (depending on rate)
- **Hurst exponent** H: 100–300 samples $\rightarrow$ 1–30 min
- **Tail index** α: 50–100 extreme samples $\rightarrow$ 5–10 min (for top 10% threshold)

Detection latency. For agents operating at 10–100 messages/minute:

- **Tier 1 (local anomaly)**: 1–10 minutes per agent
- **Tier 2–3 (spatial coherence)**: Additional 1–5 minutes for aggregation
- **Total early warning**: 2–15 minutes from swarm activation

6.3 Mechanistic link: coordination → statistical signatures

Why do coordinated swarms produce heavy tails, high kurtosis, and persistence in network traffic?

Normal (uncoordinated) agent. Independent agents generate messages stochastically:

- IMI approximately exponential (Poisson process)
- Light tails: $\alpha \approx 3$
- Memoryless: $H \approx 0.5$
- Low kurtosis: $\kappa \approx 3$ (Gaussian-like)

Coordinated swarm agent. Agents responding to coordination signals exhibit:

1. **Heavy tails (low $\alpha < 2$)**:
   - Alternates between burst phases (rapid messaging during coordinated action) and quiet phases (waiting for triggers).
   - Creates power-law-like IMI distribution with long gaps.
   - *Mechanism*: Discrete operational modes induce mixture distribution.
2. **High kurtosis ($\kappa \gg 3$)**:
   - Outlier intervals from synchronization gaps.
   - More extreme values than independent Poisson.
   - *Mechanism*: Episodic coordination events create fat-tailed deviations.
3. **Persistence (high $H > 0.6$)**:
   - Current message rate predicts future rate (bursts persist).
   - Long-range temporal dependence.
   - *Mechanism*: Coordinated state changes slowly; agents remain in "active" or "dormant" modes for extended periods.

Spatial coherence: the key differentiator. Individual agents may legitimately exhibit bursty behavior (e.g., user-triggered activity). The critical swarm signature is:

- **Spatial clustering**: Multiple agents in network proximity showing *simultaneous* anomalous patterns.
- **Temporal synchronization**: Anomalies emerge across the swarm within a narrow time window.
- **Multi-scale persistence**: Coherence strengthens or persists under coarse-graining (Tier 3).

This mechanistic justification grounds the choice of α, κ, and H as discriminative features for swarm detection.

6.4 Spatial representations in network topology

The Moran's I defines spatial weights w_{ij} that encode proximity. For P2P networks, "spatial" admits multiple interpretations:

1. **Geographic proximity**: If agent locations known (e.g., IP geolocation), use physical distance:
$$w_{ij} = \exp\left(-\frac{\|\mathbf{r}_i - \mathbf{r}_j\|^2}{2\sigma^2}\right).$$
2. **Network topology**: If communication graph available, use edge weights or graph distance:
$$w_{ij} = \begin{cases} 1/d_G(i,j) & \text{if } d_G(i,j) \leq d_{\max}, \\ 0 & \text{otherwise,} \end{cases}$$
where d_G is graph distance (hops).
3. **Hybrid**: Combine geographic and topological proximity for agents that exhibit both spatial and logical clustering.

This flexibility allows deployment across heterogeneous P2P architectures (geographic meshes, DHTs, social graphs).

7 Temporal extension

While this work establishes the feasibility of hierarchical magnetization analysis for swarm detection, temporal evolution presents fundamental challenges not addressed in the current static framework:

1. Device mobility: Moran balls at time t_0 and t_1 may represent different devices at the same spatial location (Eulerian) or the same device at different locations (Lagrangian), complicating direct comparison of local spin fields.
2. Dynamic neighborhood structure: The spatial weight matrix W(t) becomes time-dependent as peer proximity changes, meaning local spin values at different times are computed over different neighbor sets.
3. Swarm adaptation: Coordinated adversaries may employ temporal dispersal tactics, requiring pattern recognition beyond single-snapshot analysis.

8 Ethics, privacy, and data governance

Detecting swarms in P2P and device-level telemetry raises privacy concerns. We recommend:

- on-device pre-aggregation: compute Tier-1 scores locally and only transmit aggregated, privacy-preserving descriptors,
- strict minimization of data transmitted (e.g., only anomaly scores and coarse location bins),
- transparent governance and human oversight for any mitigation actions that may affect users.

9 Related work and historical context

The conceptual backbone of our framework draws from two distinct lineages:

1. the statistical physics of phase transitions and renormalization (Kadanoff, Wilson),
2. spatial autocorrelation and quantitative geography (Moran)

9.1 Kadanoff scaling

The idea of *scaling* and of laws that remain stable under changes of observation scale originated in condensed matter physics in the mid-twentieth century. In 1966, Leo P. Kadanoff introduced the "block-spin" picture of critical phenomena: to understand how a material transitions between disordered and ordered states near a critical temperature, one can group neighboring atomic spins into blocks and define a new effective spin for each block. By studying how average magnetization and correlation functions behave as the block size increases, one can infer universal scaling laws that are independent of microscopic details. Kenneth Wilson subsequently formalized this idea as the *renormalization group* (RG), showing that physical systems can be described by a flow in the space of parameters as the observation scale changes [2, 10].

In the context of AI-swarm detection, we reinterpret these ideas conceptually rather than physically. Instead of atomic spins that take discrete values ± 1, we observe multi-dimensional anomaly metrics (e.g., tail index α, kurtosis κ, and Hurst exponent H). Each agent's state is thus a point in a multi-dimensional behavioral space. These quantities do not lend themselves to the traditional Ising model's binary symmetry, but the *philosophy* of Kadanoff scaling still applies: the essential information about coherence is not in individual agents but in how their deviations aggregate and persist across scales.

Our contribution lies in defining an effective "magnetization" $M(s)$ that generalizes the concept of order parameters to multi-dimensional, non-physical data. By first mapping anomaly vectors into a continuous spin-like scalar $\phi(A) \in [-1, 1]$ and then aggregating these through spatial kernels and block averaging, we create a mathematically tractable analogue of the block-spin hierarchy for behavioral data. This allows us to study whether coordination (as measured by spin alignment) strengthens, weakens, or remains invariant under successive coarse-graining.

Importantly, we also interpret the slope of $\log M(s)$ across scales as a form of *renormalization flow* in the informational domain. Positive slopes correspond to flows toward larger-scale coherence (akin to ordering), while negative slopes correspond to decaying, short-range correlations. This adaptation makes the renormalization metaphor operational for digital swarms and establishes a scale-aware diagnostic for emergent AI collectives.

In summary, while Kadanoff work addressed thermodynamic systems, our setting is informational. The novelty is in the translation: we transfer the conceptual machinery of block averaging,

critical scaling, and flow of effective parameters into a framework for early detection of emergent, collective digital behavior.

9.2 Moran's I and the First Law of Geography

In a separate tradition, the statistic known as Moran's I [3] emerged from spatial statistics and quantitative geography. Developed to quantify the degree of spatial autocorrelation in regional data, Moran's I measures how similar values are among nearby geographic units. Formally, for observations x_i at locations i and spatial weights w_{ij} defining neighborhood relationships, the statistic is:

$$I = \frac{N}{S_0} \frac{\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}, \tag{13}$$

where $S_0 = \sum_i \sum_j w_{ij}$. Positive I indicates that neighboring locations tend to have similar values (spatial clustering), negative I indicates dissimilar neighbors (dispersion), and $I \approx 0$ indicates randomness.

Moran's I operationalizes what Waldo Tobler later famously summarized as the *First Law of Geography*: "everything is related to everything else, but near things are more related than distant things." In practice, this principle provides a natural bridge between spatial correlation and physical fields. In geography, it quantifies how regional economic activity, temperature, or population density form smooth surfaces rather than uncorrelated noise.

We borrow this spatial formalism to make the connection between statistical physics and distributed AI agents explicit. Traditional Ising models assume a fixed, low-dimensional lattice of spins, but our system is not confined to such grids: agents exist in physical and logical spaces (e.g., geolocation, network topology, communication graph). Moran's I generalizes adjacency to arbitrary weighted graphs through w_{ij}, allowing us to import spin-like interactions into non-Euclidean or irregular domains.

In our framework, the weighted local averages derived from Moran's I define what we call the *Moran-ball field*—the level-0 coarse-grained representation of local behavior. Each local field value acts as a continuous spin, suitable for input into Kadanoff-style block aggregation. This use of Moran's I as a "bridge operator" between geographic correlation and renormalization-style scaling is, to our knowledge, novel. It enables us to move from multidimensional probability metrics (tail index, kurtosis, etc.) that live in statistical space to continuous scalar fields that can be processed by hierarchical coarse-graining.

Thus, the connection between geography and statistical physics becomes operational: geography supplies the mathematical machinery (weighted neighborhood relationships) that allows us to treat irregular, real-world agent distributions as if they were embedded in a spin lattice. Physics, in turn, supplies the conceptual and mathematical language of scale transitions and emergent order. Together, they form a unified framework capable of describing emergent coordination in AI swarms that evolve in both space and time.

9.3 Novel synthesis

The novelty of our approach lies not in redefining these classical constructs but in combining them:

- We use **Moran's I** to establish spatial dependence and transform irregular, heterogeneous data into a field of continuous spin-like quantities.
- We use **Kadanoff–Wilson scaling** to explore how this field behaves under successive coarse-graining, yielding a multi-scale coherence measure interpretable as magnetization.

Together, these operations enable a conceptual *seismograph for collective AI behavior*: a method to detect early, weak coordination that becomes more apparent as we move to coarser observational scales.

In doing so, we unify traditions from geography and physics into a coherent analytic structure for digital ecosystems, bridging the gap between spatial autocorrelation and scale invariance. The result is a new way to quantify and visualize the emergence of order in AI collectives without presupposing simple, binary spins or uniform lattices, and without requiring direct observation of communication content.

9.4 Relationship to network anomaly detection

Our framework intersects with but differs from classical network security approaches:

Botnet detection. Traditional botnet detectors identify coordinated malicious hosts via:

- **Traffic volume anomalies**: Sudden spikes in packet rates.
- **Signature matching**: Known malware communication patterns.
- **Graph-based methods**: Detecting dense subgraphs in communication topology[**?**].

Our contribution: Autonomous AI swarms differ from traditional botnets in semantic richness (adaptive language models) and emergent coordination (not centrally scripted). Multi-scale coherence analysis captures hierarchical coordination patterns that simple graph metrics miss.

DDoS detection. Distributed Denial-of-Service detectors focus on:

- **Aggregate flow statistics**: Entropy, Hurst exponent of packet interarrival times.
- **Spectral analysis**: Frequency-domain signatures of periodic attacks.

Our contribution: We extend single-scale traffic analysis to multi-scale spatial coherence, enabling detection of distributed, low-intensity coordination that doesn't trigger volumetric alarms.

Peer-to-peer anomaly detection. Prior work on P2P networks addresses:

- **Churn modeling**: Normal join/leave patterns vs. Sybil attacks.
- **Routing anomalies**: Eclipse attacks, BGP hijacking.

Our contribution: We focus on *behavioral coordination* among legitimate participants whose agents exhibit collective emergent patterns, a threat vector not covered by infrastructure-level P2P security.

10 Conclusion

We presented a principled, tiered architecture for early detection of emergent AI swarms that combines per-agent anomaly scoring from network traffic statistics, local spatial smoothing via Moran's I, and hierarchical block aggregation inspired by Kadanoff scaling and seismographic practice.

The synthesis of Moran's I (spatial autocorrelation) and Kadanoff scaling for distributed AI agent monitoring is, to our knowledge, unprecedented. Existing work applies these tools separately:

- Geographic information systems use Moran's I for spatial clustering but not multi-scale analysis.
- Physics-inspired network science applies renormalization to static graph structure, not dynamic behavioral time series.
- Network anomaly detection uses temporal statistics but not spatially-weighted field representations.

Our framework unifies these traditions into a coherent early-warning architecture for the emerging threat of AI swarms.

Key contributions.

1. **Observable specification**: Network message patterns (inter-message intervals, sizes) from autonomous agents provide high-frequency, privacy-preserving data suitable for minutes-scale early warning.
2. **Mechanistic grounding**: Coordinated swarms produce heavy-tailed, high-kurtosis, persistent traffic patterns due to alternating operational modes and feedback-driven synchronization.
3. **Multi-scale coherence**: Spatial aggregation (Moran fields) followed by hierarchical coarse-graining (Kadanoff blocks) distinguishes large-scale coordination from isolated anomalies.
4. **Interpretable metrics**: Magnetization $M(s)$ and scaling slope β provide actionable diagnostics for human-in-the-loop response.

Limitations and future work. This remains a *conceptual framework* requiring empirical validation:

- **Real-world data collection**: P2P traffic from deployed autonomous agents (under strict privacy governance).
- **Threshold calibration**: ROC analysis to establish operational sensitivity/specificity tradeoffs.
- **Adversarial evaluation**: Red-team testing against evasion strategies (rate limiting, spatial dispersion, mimicry).
- **Temporal extension**: Track swarm evolution over time; handle mobile agents and dynamic topologies (Section **??**).
- **Baseline methods comparison**: Benchmark against graph anomaly detection, simple threshold rules, clustering algorithms.

Broader impact. As autonomous AI agents proliferate, frameworks for detecting emergent collective behaviors become critical infrastructure. Our approach balances:

- **Privacy**: Aggregate statistical features, not message content.
- **Interpretability**: Multi-scale metrics support explainable alerts.

- **Actionability**: Tiered architecture enables staged response (soft alerts → human review → mitigation).

We hope this blueprint catalyzes responsible research into swarm detection and informs policy discussions around distributed AI governance. The source code and synthetic experiments are openly available[6, 5] to support reproducibility and extension.

A Synthetic scenarios

Because there is no public dataset of real-world P2P traffic from LLM-enabled assistants available to the authors at this time, we validate conceptually with synthetic scenarios designed to exercise characteristic modes: *Legitimate P2P*, *Coordinated Swarm*, *Directional Swarm*, and *Mixed Population*. The scenarios were generated using a simulator of our pipeline, the code and the detailed results are available on Kaggle[5].

A.1 Generative scenarios (concept)

- **Legitimate P2P:** agents' behavioral metrics sampled from distributions matching normal operating regimes (moderate kurtosis, Hurst near 0.5, $\alpha \approx 3$). Spatially uncorrelated.
- **Coordinated Swarm:** a large fraction of agents have anomalous metrics (low α, high kurtosis, high H) spatially clustered; strong negative spin.
- **Directional Swarm:** spatial gradient; one side of domain is anomalous, the other normal. This yields low fine-scale M but higher coarse-scale M and positive slope.
- **Mixed Population:** 50/50 mix of anomalous and normal agents, producing intermediate fine-scale M and increased coarse-scale M depending on clustering.

A.2 Parameter choices

The following values were used in synthetic runs that illustrate qualitative detector behavior (not claimed to be universal):

- Domain: square $[0, 100] \times [0, 100]$.
- Number of peers: $N = 100$ per scenario.
- Spatial weight: Gaussian kernel with $\sigma = 6$ and $d_{\max} = 50$.
- Anomaly-to-spin mapping: $\phi(A) = \tanh(\gamma(A_0 - A))$ with $\gamma = 4.0$, $A_0 = 0.5$.
- Block sizes (Kadanoff aggregation): initial block size $s_0 = 12.5$ and doubling each level for $L = 4$ levels.
- Magnetization combination: $M(s) = d(s) \cdot c(s)$ where $d(s) = \text{mean}(|m_B(s) - 1|)/2$ and $c(s) = 1/(1 + \text{CV})$ (weighted CV).
- Small numerical floor: $\epsilon = 10^{-6}$ when taking logs for slope estimation.

Table 1: Detector outputs (synthetic runs). M_0 is fine-scale magnetization, $M_{\max}$ is primary magnetization (maximum across scales), and β is the log-profile slope.

Scenario	M_0 (L0)	$M_{\max}$	β (slope)
Legitimate P2P	0.126	0.129	0.008
Coordinated Swarm	0.491	0.673	0.107
Directional Swarm	0.190	0.386	0.214
Mixed Population	0.211	0.439	0.227

A.3 Results

Table 1 summarizes outputs from these synthetic experiments (values are illustrative; refer to code for full logging).

A.4 Interpretation and justification of parameters

Why $\sigma = 6$? In a 100×100 domain $\sigma = 25$ overly smooths and collapses neighborhood distinctions; $\sigma \approx 6$ captures local neighborhoods roughly aligned with the initial block size of 12.5 (about two σ), preserving L0 structure while enabling meaningful L1 aggregation.

Why multiplicative $M(s) = d \cdot c$? A multiplicative combination ensures a high magnetization requires both substantial deviation from the ambient state and inter-block coherence. Additive rules can produce high M for high deviation but incoherent mixtures (false positives); multiplicative rules suppress such cases.

Why continuous spin map? Hard thresholding discards useful marginal information and can produce all-or-nothing uniform fields. A continuous tanh map preserves gradients and yields realistic, scale-sensitive behavior.

Why block-weighting? Weighting block contributions by population reduces the influence of tiny noisy blocks and ensures M reflects macroscopic structure.

A.5 Limitations of synthetic results

Synthetic scenarios are necessarily simplified: real-world agents have richer heterogeneity (device types, temporal cycles, background noise) and adversarial actors may adapt. Thus thresholds and parameters require empirical calibration on real traces (collected under strict privacy controls).

B Practical next steps for empirical work

1. Collect representative on-device telemetry under strong privacy and legal safeguards (sampled, aggregated, opt-in).
2. Implement the tiered pipeline with on-device Tier-1 scoring and cloud-side Tier-3 aggregation of privacy-preserving descriptors.
3. Calibrate thresholds using a mix of benign operational data and defensive red-team simulations.
4. Study adversarial adaptation (agents intentionally modify local behavior to avoid detection).
5. Operationalize human-in-the-loop review and graduated remediation (rate-limiting, challenge-response, targeted investigation).

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